Industrial Security (Bachelor) – Delivery Schedule

5 days, ∼40 hours. 10 sections, 211 slides, 100 exercises, 10 hands-on labs. Timing is a guide – adjust to room dynamics.

Day 1 – Foundations

Slot	Section	Slides	Exercises	Lab	Watch for
09:00–10:30	01 – Introduction	Frames 1–13, ∼90 min	1.1–1.5 (after slides)	—	The IT/OT contrast question is the productive one. Mine it.
10:30–10:45	*Break*				
10:45–12:15	01 cont. + 02 – ICS Fundamentals	01 wrap + 02 frames 1–7	1.6–1.10 + 2.1–2.3	—	Scan cycle is load-bearing for everything later. Don't rush it.
13:30–15:00	02 – ICS Fundamentals cont.	02 frames 8–13	2.4–2.7	—	Ladder logic: walk one rung live on the board.
15:00–17:00	Lab 02 – Stand up the Stack	—	2.8–2.10 (parallel)	`02-ics-fundamentals` (Docker stack up; first Modbus read)	First lab: budget time for environment setup.

Day 2 – Network and Protocols

Slot	Section	Slides	Exercises	Lab	Watch for
09:00–10:30	03 – Network Architecture	Frames 1–7, Purdue → zones & conduits	3.1–3.4	—	Purdue is conventional, not physical – say so on slide 1.
10:45–12:15	03 cont.	Frames 8–13, firewalls / data diodes / wireless	3.5–3.10	—	VLAN ≠ segmentation; force them to write the ACL.
13:30–15:00	04 – Industrial Protocols	Frames 1–7, Modbus → EtherNet/IP	4.1–4.5	—	Decode the Modbus frame on the board; it's the load-bearing trick.
15:00–17:00	04 cont. + Lab 04	Frames 8–12, OPC UA / MQTT / 61850	4.6–4.10 (during lab)	`04-protocols` (Modbus inject, OPC UA browse, PCAP diff)	The injection step is the "aha"; let them run it themselves.

Day 3 – Threats and Vulnerabilities (the heaviest day)

Slot	Section	Slides	Exercises	Lab	Watch for
09:00–10:30	05 – Threat Landscape	Actors + ATT&CK + Stuxnet (3 frames)	5.1 during slides	—	Read the matrix left-to-right; map every case study back to it.
10:45–12:15	05 cont. – Case Studies	Ukraine 15/16, TRITON, Industroyer	5.2–5.3	—	2015 vs. 2016 contrast is the lesson; the rest is colour.
13:30–15:00	05 cont. – Case Studies	NotPetya, Colonial, PIPEDREAM, CyberAv3ngers, Volt Typhoon	5.4–5.10	—	PIPEDREAM "caught before victim" is the geopolitical point.
15:00–17:00	06 – Vulnerabilities + Lab 06	Frames 1–13 (compact)	6.1–6.10 (during lab)	`06-vulnerabilities` (asset inventory, safe nmap)	The 200-PLC matching exercise is the keeper.

Day 4 – Risk and Defence

Slot	Section	Slides	Exercises	Lab	Watch for
09:00–10:30	07 – Risk Management	Frames 1–7, risk matrix → Bow-Tie	7.1–7.3	—	Risk matrix colour bands are policy, not truth. Repeat.
10:45–12:15	07 cont.	Frames 8–14, LOPA / CCE / FAIR / insurance	7.4–7.10	—	Walk one LOPA row live with PFDs.
13:30–15:00	08 – Defense in Depth	Frames 1–7, onion / hardening / remote access	8.1–8.5	—	"Three layers, one logo" is the misconception to attack.
15:00–17:00	08 cont. + Lab 08	Frames 8–13, NAC / IDS / SIEM / deception	8.6–8.10 (during lab)	`08-defense-in-depth` (Zeek alerts on Modbus traffic)	First time most students see a real Zeek log.

Day 5 – Standards and Incident Response

Slot	Section	Slides	Exercises	Lab	Watch for
09:00–10:30	09 – Standards & Governance	Frames 1–7, 62443 / NIST / NIS2 / KRITIS	9.1–9.5	—	Two dates: NIS2 17 Oct 2024; CRA 11 Dec 2027.
10:45–12:15	09 cont.	Frames 8–13, CRA / ENISA / C2M2 / programme	9.6–9.10	—	"Compliance $\neq$ security" is the slogan they'll quote later.
13:30–15:00	10 – Incident Response	Frames 1–7, OT IR differences $\rightarrow$ forensics	10.1–10.5	—	Pulling the cable in OT can trip SIS. Anchor this.
15:00–16:30	10 cont. + Lab 10	Frames 8–13, tabletop / BCP / post-mortem	10.6–10.10 (during lab)	`10-incident-response` (tabletop walkthrough on the stack)	Tabletop runs better with you facilitating, not lecturing.
16:30–17:00	Wrap-up	Recap takeaway slides per section	—	—	5-minute Q&A; thank students; share contact for follow-ups.

Pre-flight check (run the morning before Day 1): `bachelor/labs/preflight.sh --pull --bootstrap`. All 16 checks must pass: services up, landing page at `http://127.0.0.1:8000`, OpenPLC primed with `conveyor.st`, Modbus 502 open, OPC UA 4840 open, Zeek raises the `ModbusWatch::Unsolicited_Write` notice on injection. Each lecturer machine needs: Docker $\geq$ 24, $\sim$3 GB free for images, host ports 8000 (landing), 8080 (OpenPLC UI), 502, 4840, 1883 free, and projector-readable terminal font ($\geq$ 18 pt).

Per-slide artefacts you can lean on: the `NN-topic-notes.pdf` alongside each slide deck shows your prepared notes on the right; project the `-slides.pdf` only. Each section ends with a *Key Takeaway* frame – that is your natural cliff-hanger into the next section. The *Section Roadmap* on the first frame of Section 01 is a useful re-orient point if students get lost mid-day.